

PROBLEM STATEMENT #12 | Healthcare & Telemedicine**Blood Bank Inventory & Emergency Donor Matcher****1. Problem Context & Overview**

Blood banks require a real-time stock management system that monitors blood component shelf lives and triggers geo-targeted emergency notifications to matching eligible donors during critical shortages.

Target Stakeholders / Actors: *Emergency Requester, Blood Bank Manager*

2. Sample Functional Requirement (FR) Guideline**Sample Requirement: FR-001 [Priority: High]**

Description: The system shall cross-reference emergency blood unit requests against real-time blood bank inventory and broadcast SMS alerts to compatible donors within a 10 km radius.

Sample Acceptance Criteria: Pass: Compatible donors notified within 30 seconds of emergency flag; Fail: Incompatible blood group alerted.

3. Sample Non-Functional Requirement (NFR) Guideline**Sample Requirement: NFR-001 [Type: Performance & Security]**

Description: The inventory ledger must maintain strict transactional consistency, preventing simultaneous allocation of the same blood bag.

Sample Acceptance Criteria: Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.

Deliverables to Submit: [All through a GitHub Repo in your account]**• Complete Requirements Table:**

- >> Exactly 5 Functional Requirements (FR-001 to FR-005)
- >> 2 Non-Functional Requirements (NFR-001 & NFR-002) with ID, Type, Description, Priority, Acceptance Criteria, and Rationale.

• UML Use-Case Diagram:

- >> Model all actors, primary use cases, and include at least one «include» and one «extend» relationship.

• Use-Case Flow Specification:

- >> 1-page document for one core use case detailing Preconditions, Postconditions, Main Success Scenario
- >> One Alternate Flow.